STATE of the ART

Reading Comprehension Tasks Requiring Logical Reasoning, the following papers are taking the advantages of logical equivalence, and common inference for improving logical reasoning for language models.

- [1] Logic-driven data augmentation
- [2] Intermediate training / Continual Pre-training
- [3] Evaluation for more challenged and robust logical reasoning tasks
- [4] Mathematical programming solver/optimizer

Paper title	Key points	Dataset	Paper link
[1] [2] Abstract Meaning Representation-Based Logic-Driven Data Augmentation for Logical Reasoning	It presents an AMR-based logic-driven data augmentation method for training and prompting large language models and performing SOTA during that time.	ReClor LogiQA	https://arxiv.org/abs/2305.12599
[3] Assessing and Enhancing the Robustness of Large Language Models with Task Structure Variations for Logical Reasoning	It presents a set of challenged dataset for evaluating model's performance on complex logical reasoning tasks	ReClor LogiQA LogiQAv2	https://arxiv.org/abs/2310.09430
[4] MindOpt (Machine Intelligence of Damo OPTimizer), a high performance mathematical programming optimizer	https://github.com/aliyun/mindoptj	ReClor (Currently they are standing at the top of ReClor leaderboard)	No paper at the moment
[1] [2] IDOL: Indicator-oriented Logic Pre-training for Logical Reasoning	https://github.com/GeekDream-x/IDOL Intermediate training with constructed indicator-oriented training strategy	ReClor (#1 before AMR-LDA) LogiQA	https://arxiv.org/abs/2306.15273
[1] [2] MERIt: Meta-Path Guided Contrastive Learning for Logical Reasoning	https://github.com/SparkJiao/MERIt Intermediate training with constructed reading comprehension intermediate training	ReClor (#1 before IDOL) LogiQA	https://arxiv.org/abs/2203.00357
[1] Logic-Driven Context Extension and	https://github.com/SiyuanWangw/LReasoner	ReClor (#1 before MERIt)	https://arxiv.org/abs/2105.03659

Data Augmentation for Logical Reasoning of Text	Syntax based rule and template for logic-driven data augmentation	LogiQA	
The Reversal Curse: LLMs trained on "A is B" fail to learn "B is A"	Reversal Curse.	1.NameToDescription pair. 2.1573 child-parent relations.	https://arxiv.org/abs/2309.12288
Does Refusal Training in LLMs Generalize to the Past Tense?	Past tense attack still struggling for the existing front-tier llms.	JBB-Behaviors past tense examples d	https://arxiv.org/abs/2407.11969
[Premise order matters in reasoning with large language models]()	Premise order matters in deductive reasoning (similar to our previous findings in Multi-Step Deductive Reasoning Over Natural Language: An Empirical Study on Out-of-Distribution Generalisation)	R-GSM	https://arxiv.org/abs/2402.08939
Dyval: Dynamic evaluation of large language models for reasoning tasks	Avoid large language model memory the benchmark, relying on the graph structure to generate samples	https://github.com/microsoft/promptbench	https://openreview.net/forum?id=gjfOL9z5Xr
Dyval2: Dynamic Evaluation of Large Language Models by Meta Probing Agents	multifaceted analysis, LLM-based agents to automatically transform existing problems into new ones, which are more flexible and support diverse tasks.	https://github.com/microsoft/promptbench	https://arxiv.org/pdf/2402.14865

NEW papers:

Reverse Training to Nurse the Reversal Curse	https://arxiv.org/abs/2403.13799		
Models Can and Should Embrace the Communicative Nature of Human-Generated Math	https://arxiv.org/pdf/2409.17005		
Physics of Language Models: Part 3.2, Knowledge Manipulation	https://arxiv.org/pdf/2309.14402		
Towards a Theoretical Understanding of the 'Reversal Curse' via Training Dynamics	https://arxiv.org/abs/2405.04669		
Supposedly Equivalent Facts That Aren't? Entity Frequency in Pre-training Induces Asymmetry in LLMs	https://arxiv.org/pdf/2503.22362		
Is the Reversal Curse a Binding Problem? Uncovering Limitations of Transformers from a Basic Generalization Failure	https://arxiv.org/pdf/2504.01928 https://github.com/neobundy/Deep-Dive-Into-AI-With-MLX-PyTorch/blob/main/deep-dives/015-meta-jepa/README.md		
Order Doesn't Matter, But Reasoning Does: Training LLMs with Order-Centric Augmentation	https://arxiv.org/pdf/2502.19907		
Understanding the Logical Capabilities of Large Language Models via	https://arxiv.org/pdf/2503.10408		

Out-of-Context Representation Learning			
Beyond Chain-of-Thought, Effective Graph-of-Thought Reasoning in Language Models	https://arxiv.org/pdf/2305.16582	https://github.com/Zoeyyao27/Graph-of-Thought	
DAGN: Discourse-Aware Graph Network for Logical Reasoning	https://arxiv.org/pdf/2103.14349		
Graph of Thoughts: Solving Elaborate Problems with Large Language Models	https://arxiv.org/abs/2308.09687		
Graph Neural Prompting with Large Language Models	https://arxiv.org/abs/2309.15427		
KG-CoT: Chain-of-Thought Prompting of Large Language Models over Knowledge Graphs for Knowledge-Aware Question Answering	https://www.ijcai.org/proceedings/2024/734		

Mitigating Reversal Curse in Large Language Models via Semantic-aware Permutation Training	https://arxiv.org/pdf/2403.00758

LIMA: Less Is More for Alignment	https://arxiv.org/pdf/2305.11206
LIMO: Less is More for Reasoning	https://arxiv.org/pdf/2502.03387v1
Beyond Chain-of-Thought, Effective Graph-of-Thought Reasoning in Language Models	https://arxiv.org/abs/2305.16582 ttps://github.com/Zoeyyao27/Graph-of-Thought
DAGN:Discourse-Aware Graph Network for Logical Reasoning	https://arxiv.org/abs/2103.14349
Explaining Answers with Entailment Trees	https://aclanthology.org/2021.emnlp-main.585/
ProofWriter: Generating Implications, Proofs, and Abductive Statements over Natural Language	https://arxiv.org/abs/2012.13048
Graph Chain-of-Thought: Augmenting Large Language Models by Reasoning on Graphs	https://arxiv.org/pdf/2404.07103
Empowering llms with logical reasoning: A comprehensive survey	https://arxiv.org/abs/2502.15652
Enhancing Reasoning Capabilities of LLMs via Principled Synthetic Logic Corpus	https://arxiv.org/abs/2411.12498
Logical Consistency of Large Language Models in Fact-checking	https://arxiv.org/abs/2412.16100

Candidate models
Equivariant neural network
https://arxiv.org/abs/2205.07362

Some interesting findings from Cycorp
https://www.linkedin.com/posts/cycorp_enhancing-llm-performance-with-cyc-knowledge-activity-7206394462929846272-0PZ8?utm_source=share&utm_medium=member_desktop

Example of using Cyc to make ChatGPT smarter
https://chatgpt.com/share/ab7f7da9-c23e-49f5-8974-849a63b09f40

Research Questions

1. Existing large language models such as GPT-4 fail in the situation that A is B is equivalent to B is A. How to help large language models address this challenge?
2. Although there is large scale data on the internet, that data lacks the reasoning process to get the answer. How to automatically construct high quality data for logical reasoning and particularly for addressing A is B and B is A type of question?
3. Whether the presented logical reasoning dataset can be used to help large language models increase generalization performance on logical reasoning tasks?

Experiment Design

For the research question 1:

We can present a logic-driven data augmentation method which is going to paraphrase sentences using logical equivalent laws to augment the downstream task such as A is B is equivalent to B is A.

Existing method: LReasoner(Syntax-based logic-driven data augmentation method), AMR-LDA(Semantic-based logic-driven data augmentation method), and AMR-DA(Semantic data augmentation method), CycL-based prompt using a large language model.

For the research question 2:

Instead of only constructing the logical equivalent sentence directly to downstream tasks, can we present a method which can construct an intermediate reasoning process during constructing the finalized logical equivalent sentence and then help a large language model address A is B and B is A question?

For the research question 3:

Apply the intermediate reasoning process data augmentation pipeline to augment data and apply for different downstream reasoning tasks to see whether it can provide help.